



Department of Computer Science
University of Wah
Assignment #02

Course Code: CS-251

Course Title: Intro. to Information Security

Total Marks: 15

Discipline/Semester: BS(CS)-3rd B

Deadline: 27th Nov, 2025

Assigned Date: 20th Nov, 2025

[CLO2-GA3- C3]

Instructions:

- Submit your assignment in hard form in a well-structured format

Question 1: As you are familiar with substitution techniques in cryptography, considering these techniques. **Apply** each step of algorithms and answer the following:

- a. Using the Hill Cipher and the following 3×3 key matrix:

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

- Encrypt the plaintext "**INFORMATION SECURITY**"

Convert the resulting vectors back into letters to form the ciphertext.

- Decrypt the resulting cipher text back to the original plaintext using the inverse of the key matrix.

- b. Encrypt the binary plaintext "**SECURITY IS POWER**" using the Vernam cipher with the binary key "**KNOWLEDGE IS KEY!**"

- Show all steps of encryption clearly
- Then, decrypt the ciphertext using the same key to retrieve the original plaintext.

"Small changes can lead to big impacts—reduce, reuse, and save energy for a brighter future."